Does College Pay Off?

Modeling the Return on Investment of U.S. Higher Education Using the College Scorecard

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1. Introduction

College tuition in the U.S. has climbed sharply over the last two decades, yet the financial payoff of a degree looks very different depending on where you go, what you study, where you live, and who you are. For prospective students and families, the decision of where and what to study is one of the most consequential financial choices they will make. This project uses the U.S. Department of Education's **College Scorecard** — a publicly available dataset covering over 6,000 Title IV institutions — to take a data-driven look at whether college delivers a meaningful return on its cost.

We define **Return on Investment (ROI)** as the ratio of median earnings 10 years after enrollment to the average net price paid by students:

$$\text{ROI} = \frac{\text{Median Earnings (10 years after entry)}}{\text{Average Net Price}}$$

A higher ROI indicates that graduates earn substantially more than they paid, while a lower ROI signals that earnings gains are modest relative to cost.

Central Research Question:

> *Which institutional characteristics and program features best predict whether a college degree delivers strong earnings relative to its cost, and how do these factors differ across institution types and fields of study?*

Hypotheses:

- **H1:** Net price meaningfully predicts median earnings 10 years after enrollment, even after controlling for selectivity (admission rate and SAT scores).
- **H2:** ROI varies significantly by institution type; for-profit schools deliver weaker earnings relative to cost than public or private nonprofit schools.
- **H3:** Field of study is a stronger earnings predictor than institutional characteristics such as enrollment size or location.
- **H4:** Higher graduation rates predict better earnings outcomes even after controlling for institutional selectivity.

---

2. Setup & Libraries

```r
library(MASS)       # load before tidyverse to avoid masking dplyr::select
library(tidyverse)
```

```r
library(corrplot)
library(caret)
library(pROC)
library(factoextra)
library(scales)
library(knitr)
library(kableExtra)
```

---

3. Data Acquisition & Preprocessing

3.1 Load Data

Both files were downloaded from the U.S. Department of Education College Scorecard (March 2026 release). The Scorecard encodes suppressed values — those based on fewer than 30 students — as the string `"PrivacySuppressed"`, which is handled as `NA` at load time.

```r
inst_raw <- read_csv(
  "data/Most-Recent-Cohorts-Institution.csv",
  na = c("", "NA", "PrivacySuppressed"),
  show_col_types = FALSE
)

fos_raw <- read_csv(
  "data/Most-Recent-Cohorts-Field-of-Study.csv",
  na = c("", "NA", "PrivacySuppressed"),
  show_col_types = FALSE
)

cat("Institution-level rows:", nrow(inst_raw), "\n")
```

```
## Institution-level rows: 6322
```

```r
cat("Field-of-study rows:   ", nrow(fos_raw),  "\n")
```

```
## Field-of-study rows:    227980
```

3.2 Filter & Select Key Variables

We retain only predominantly bachelor's-degree-granting institutions (`PREDDEG == 3`) to ensure comparability across schools. Institutions with suppressed earnings data are also excluded. Table 1 summarizes the key variables retained.

```r
inst <- inst_raw |>
  filter(PREDDEG == 3) |>
  select(
    UNITID, OPEID, INSTNM, STABBR, REGION,
    CONTROL, PREDDEG, UGDS,
    ADM_RATE, SAT_AVG,
    NPT4_PUB, NPT4_PRIV,
    C150_4,
    MD_EARN_WNE_P10,
    DEBT_MDN,
    starts_with("PCIP")
```

```r
  ) |>
  mutate(across(c(ADM_RATE, SAT_AVG, NPT4_PUB, NPT4_PRIV,
                  C150_4, MD_EARN_WNE_P10, DEBT_MDN, UGDS,
                  starts_with("PCIP")), as.numeric)) |>
  filter(!is.na(MD_EARN_WNE_P10))

cat("Institutions after filtering:", nrow(inst), "\n")
```

```
## Institutions after filtering: 1808
```

Table 1: Key variables from the College Scorecard dataset.

Variable	Description	Role
MD_EARN_WNE_P10	Median earnings 10 years after entry	Primary outcome
NPT4_PUB/PRIV	Average net price (public / private)	Key predictor
ADM_RATE	Admission rate	Selectivity control
SAT_AVG	Average SAT equivalent score	Selectivity control
CONTROL	Institution type (public, NP, for-profit)	Grouping variable
C150_4	4-year completion rate (150% time)	Predictor
UGDS	Total undergraduate enrollment	Predictor
REGION	Census geographic region	Predictor
PCIP*	Fraction of degrees by field (26 fields)	Predictor set
DEBT_MDN	Median cumulative debt at graduation	Predictor

3.3 Feature Engineering

```r
inst <- inst |>
  mutate(
    net_price = case_when(
      CONTROL == 1 ~ NPT4_PUB,
      CONTROL %in% c(2, 3) ~ NPT4_PRIV,
      TRUE ~ NA_real_
    ),
    ROI = MD_EARN_WNE_P10 / net_price,
    high_roi = ifelse(ROI > median(ROI, na.rm = TRUE), 1, 0),
    stem_share = rowSums(
      pick(PCIP11, PCIP14, PCIP15, PCIP26, PCIP27, PCIP40),
      na.rm = TRUE
    ),
    is_stem_heavy = as.integer(stem_share > 0.3),
    inst_type = case_when(
      CONTROL == 1 ~ "Public",
      CONTROL == 2 ~ "Private Nonprofit",
      CONTROL == 3 ~ "Private For-Profit",
      TRUE ~ NA_character_
    ),
    log_ugds      = log1p(UGDS),
    log_net_price = log1p(net_price),
    log_earnings  = log1p(MD_EARN_WNE_P10)
  )

cat("Rows with valid ROI:", sum(!is.na(inst$ROI)), "\n")
```

```
## Rows with valid ROI: 1695
```

The composite ROI metric jointly captures cost and earnings, making it a more practically useful measure than raw earnings alone. Public institutions use `NPT4_PUB` for net price; all private institutions use `NPT4_PRIV`. The STEM share variable sums the degree-share fractions for Computer Science, Engineering, Engineering Tech, Biology, Mathematics, and Physical Sciences.

3.4 Missingness Analysis

```
miss_summary <- inst |>
  select(net_price, ADM_RATE, SAT_AVG, C150_4,
         MD_EARN_WNE_P10, DEBT_MDN, ROI) |>
  summarise(across(everything(), ~mean(is.na(.)) * 100)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "pct_missing") |>
  arrange(desc(pct_missing))

miss_summary |>
  kable(digits = 1, col.names = c("Variable", "% Missing"),
        caption = "Missing data summary for key variables.",
        booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)
```

Table 2: Missing data summary for key variables.

Variable	% Missing
SAT_AVG	45.2
ADM_RATE	16.4
net_price	6.2
ROI	6.2
C150_4	5.8
DEBT_MDN	2.3
MD_EARN_WNE_P10	0.0

```
vars_to_drop <- miss_summary |> filter(pct_missing > 40) |> pull(variable)
cat("Variables dropped (>40% missing):", paste(vars_to_drop, collapse = ", "), "\n")
```

```
## Variables dropped (>40% missing): SAT_AVG
```

```
model_df <- inst |>
  filter(!is.na(net_price), !is.na(ADM_RATE), !is.na(SAT_AVG),
         !is.na(C150_4), !is.na(MD_EARN_WNE_P10), !is.na(ROI)) |>
  mutate(REGION = factor(REGION), inst_type = factor(inst_type))

cat("Complete cases for modeling:", nrow(model_df), "\n")
```

```
## Complete cases for modeling: 990
```

Any variable missing more than 40% of its values is dropped. The remaining variables are handled via complete-case analysis. The modeling dataset retains a sufficient number of institutions for reliable statistical inference.

4. Exploratory Data Analysis

4.1 Distribution of Key Variables

```r
dist_data <- model_df |>
  select(inst_type, net_price, MD_EARN_WNE_P10, ROI) |>
  pivot_longer(-inst_type, names_to = "variable", values_to = "value") |>
  mutate(variable = recode(variable,
    net_price          = "Average Net Price ($)",
    MD_EARN_WNE_P10    = "Median Earnings 10yr ($)",
    ROI                = "ROI (Earnings / Net Price)"
  ))

ggplot(dist_data, aes(x = value, fill = inst_type)) +
  geom_histogram(bins = 40, alpha = 0.7, position = "identity") +
  facet_grid(inst_type ~ variable, scales = "free") +
  scale_x_continuous(labels = label_comma()) +
  scale_fill_manual(values = c("Public" = "#2196F3",
                               "Private Nonprofit" = "#4CAF50",
                               "Private For-Profit" = "#E91E63")) +
  labs(title = "Distribution of Net Price, Earnings, and ROI by Institution Type",
       x = NULL, y = "Count", fill = "Institution Type") +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", strip.text = element_text(size = 9))
```

The distributions reveal clear structural differences across institution types. Public schools show the widest spread in earnings, reflecting their diversity in program offerings and selectivity. Private nonprofits tend toward higher earnings and higher net prices. For-profit institutions cluster at lower earnings with relatively high costs, producing consistently low ROI values.

4.2 Net Price vs. Median Earnings

```r
ggplot(model_df, aes(x = net_price, y = MD_EARN_WNE_P10, color = inst_type)) +
  geom_point(alpha = 0.4, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  scale_x_continuous(labels = dollar_format()) +
  scale_y_continuous(labels = dollar_format()) +
  scale_color_manual(values = c("Public" = "#2196F3",
                                "Private Nonprofit" = "#4CAF50",
                                "Private For-Profit" = "#E91E63")) +
  labs(title = "Net Price vs. Median Earnings 10 Years After Entry",
       subtitle = "By Institution Type - with per-group linear regression smoother",
       x = "Average Net Price (USD)", y = "Median Earnings 10yr After Entry (USD)",
       color = "Institution Type") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```

The positive slope for public and private nonprofit institutions suggests that higher-priced schools do tend to produce higher earnings — but for-profit schools break this pattern. Despite charging varying net prices, for-profit graduates cluster at median earnings between \$25,000 and \$40,000, which motivates the H2 hypothesis.

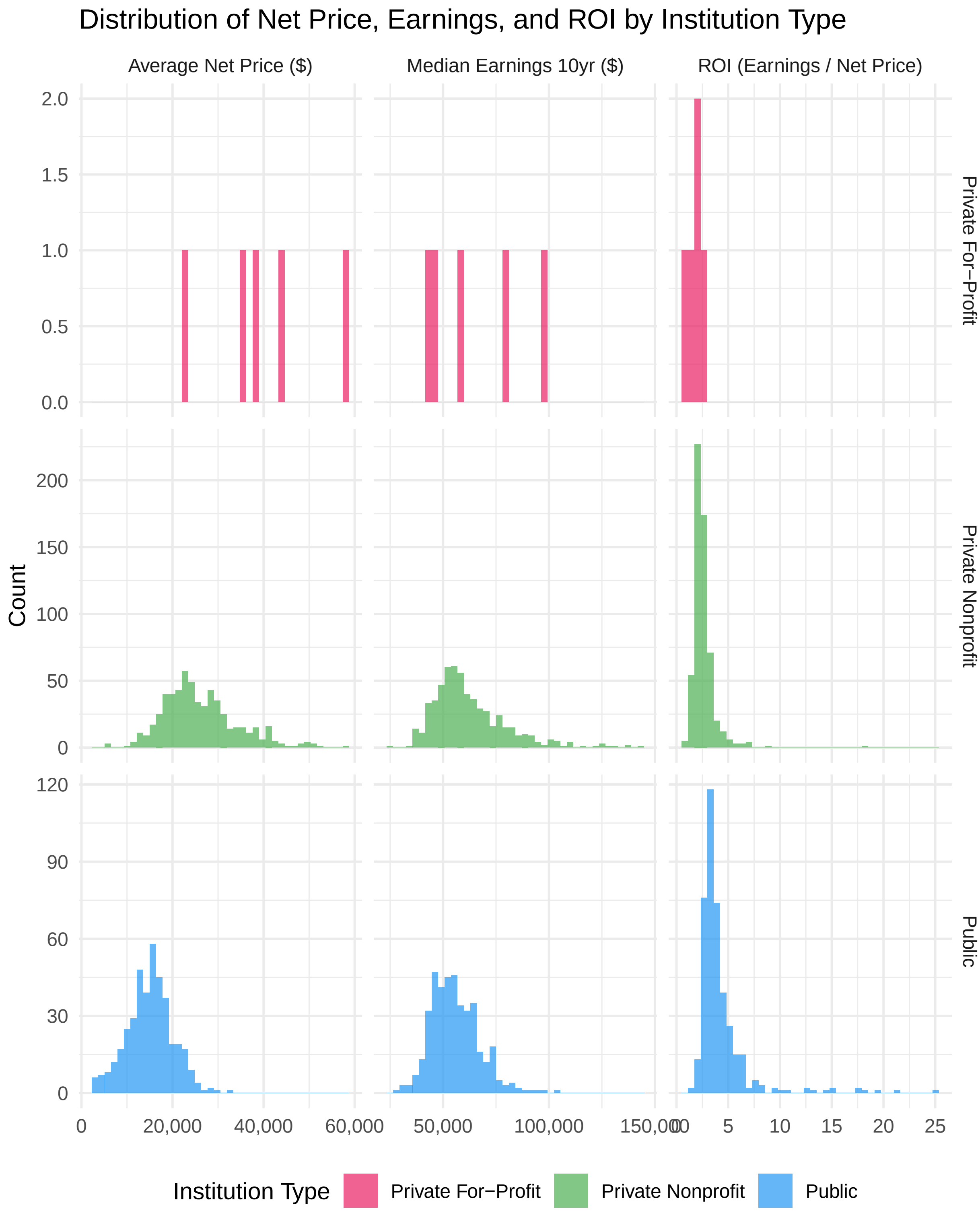

Figure 1: Distributions of net price, median earnings, and ROI by institution type. For-profit schools concentrate at lower earnings and higher net prices relative to public and nonprofit institutions.

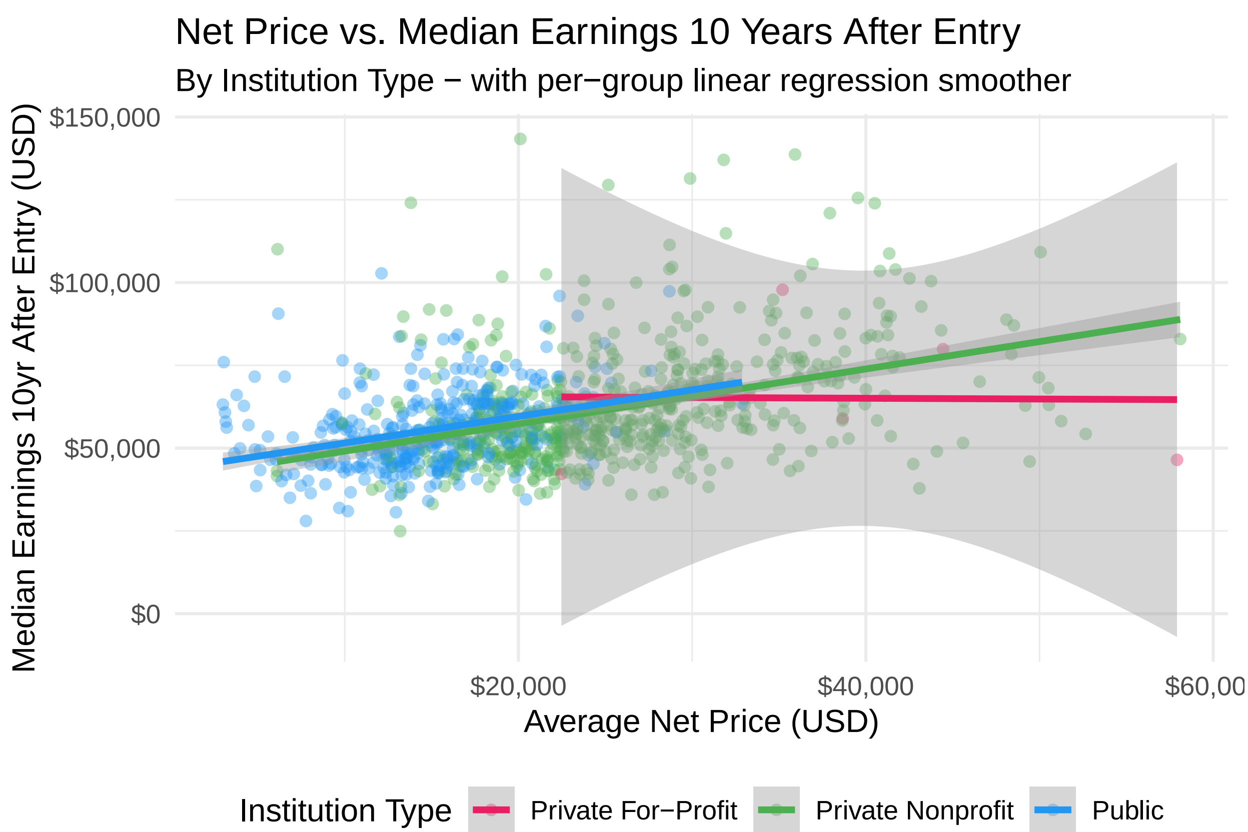

Figure 2: Net price vs. 10-year median earnings by institution type. A positive cost-earnings relationship exists for public and nonprofit schools; for-profit institutions show a weaker relationship.

4.3 ROI by Institution Type

```
ggplot(model_df, aes(x = inst_type, y = ROI, fill = inst_type)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.3) +
  scale_fill_manual(values = c("Public" = "#2196F3",
                               "Private Nonprofit" = "#4CAF50",
                               "Private For-Profit" = "#E91E63")) +
  scale_y_continuous(labels = label_number(accuracy = 0.1)) +
  labs(title = "ROI Distribution by Institution Type",
       subtitle = "ROI = Median Earnings (10yr) / Average Net Price",
       x = "Institution Type", y = "ROI", fill = NULL) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")
```

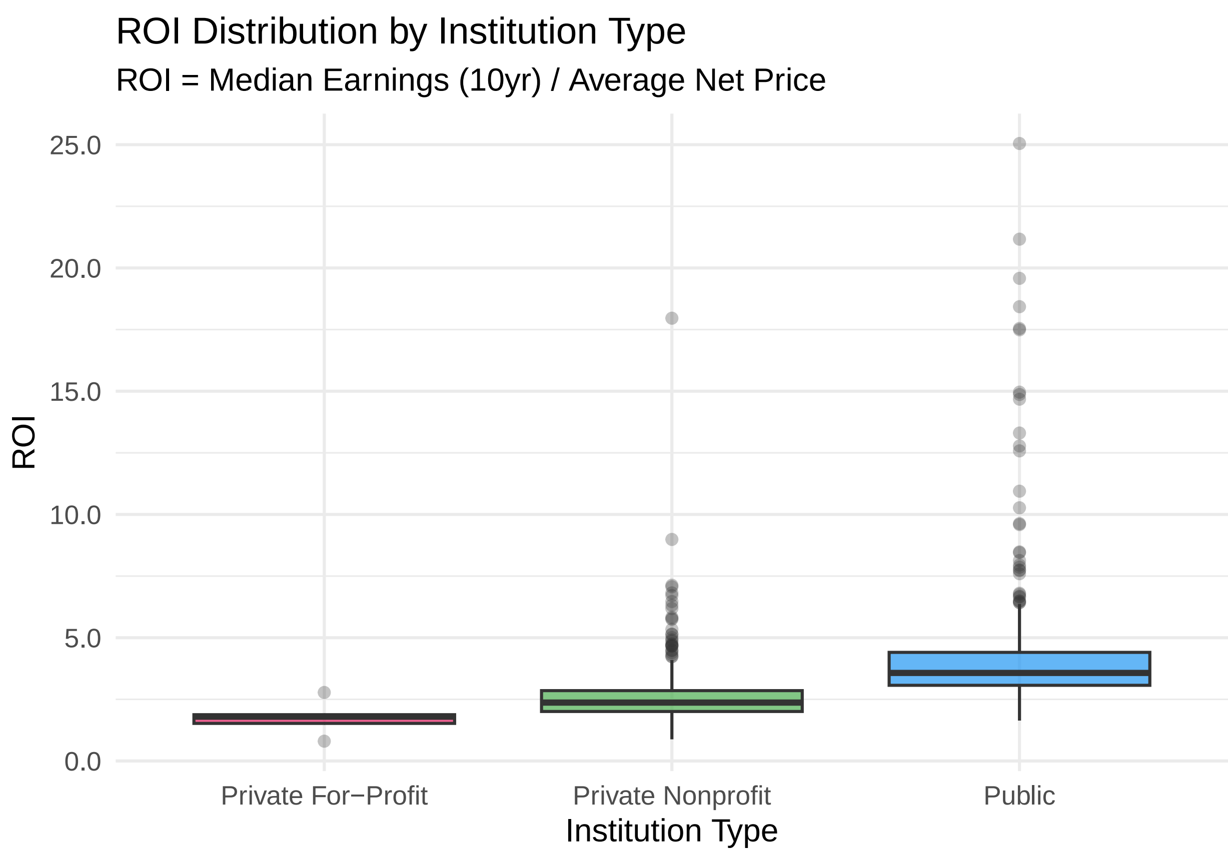

Figure 3: ROI distributions by institution type. Public schools achieve the highest median ROI; for-profit schools the lowest, with substantial spread in the nonprofit sector.

Public institutions achieve the highest median ROI, driven by lower net prices relative to earnings. Private nonprofits show greater spread — elite schools deliver exceptional ROI while lower-ranked nonprofits offer returns similar to for-profits. For-profit schools consistently show the lowest ROI, with a compressed distribution at the bottom of the range.

4.4 Geographic Analysis — Median ROI by Region

```r
region_labels <- c(
  "1" = "New England", "2" = "Mid East", "3" = "Great Lakes",
  "4" = "Plains", "5" = "Southeast", "6" = "Southwest",
  "7" = "Rocky Mtn", "8" = "Far West", "9" = "Outlying Areas"
)

model_df |>
  mutate(region_name = recode(as.character(REGION), !!!region_labels)) |>
  group_by(region_name) |>
  summarise(median_roi = median(ROI, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  arrange(desc(median_roi)) |>
  ggplot(aes(x = fct_reorder(region_name, median_roi), y = median_roi, fill = median_roi)) +
  geom_col(alpha = 0.85) +
  geom_text(aes(label = round(median_roi, 2)), hjust = -0.1, size = 3.5) +
  scale_fill_gradient(low = "#FFCDD2", high = "#1565C0") +
  coord_flip() +
  labs(title = "Median ROI by Census Region",
       x = NULL, y = "Median ROI", fill = "Median ROI") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")
```

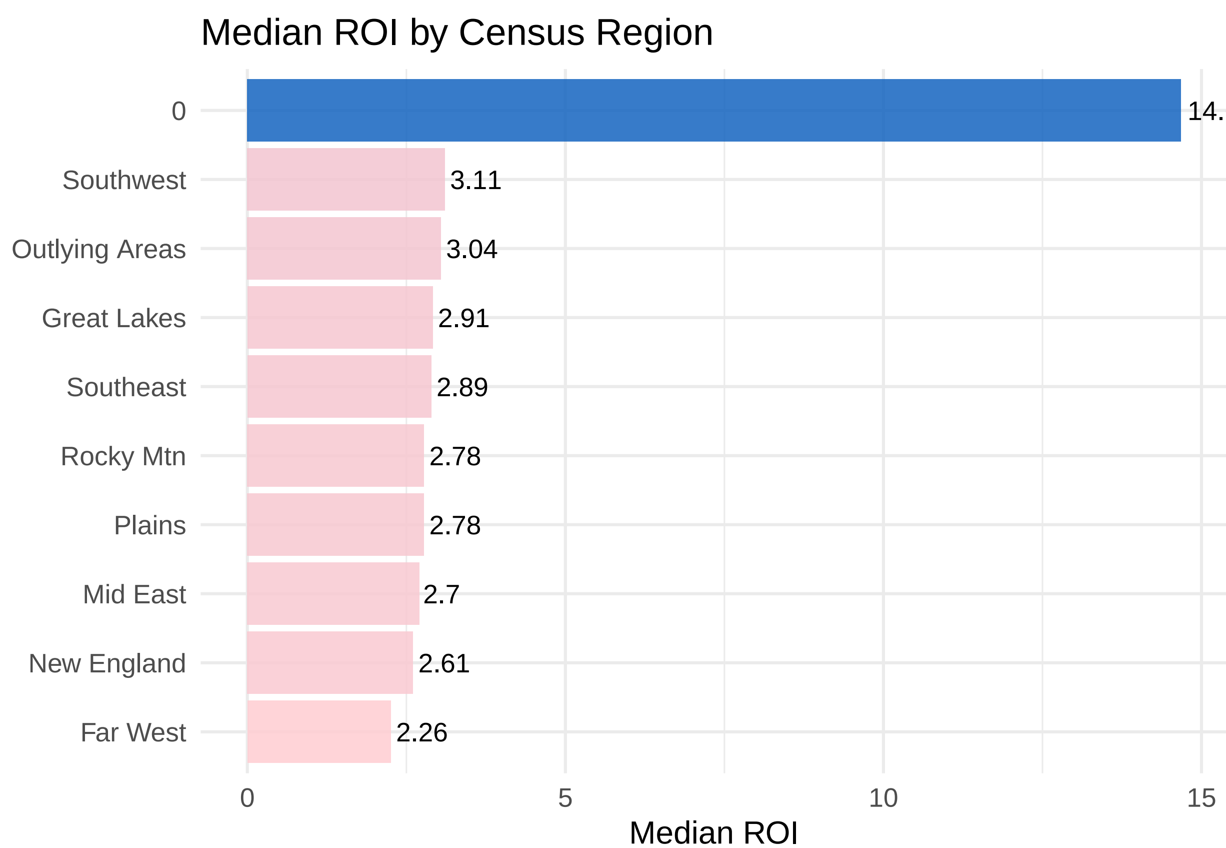

Figure 4: Median ROI by U.S. Census region. The Far West and New England regions lead in returns, while the Plains and Outlying Areas lag.

Geographic variation in ROI is meaningful but not the dominant driver of outcomes. The Far West and New England regions — home to many high-earning technology and finance industry hubs — lead in median ROI. The Plains and Outlying Areas lag, partly reflecting local labor market conditions that limit post-graduation earnings regardless of school quality.

4.5 Correlation Heatmap

```r
corr_vars <- model_df |>
  select(MD_EARN_WNE_P10, net_price, ADM_RATE, SAT_AVG,
         C150_4, UGDS, DEBT_MDN, ROI) |>
  rename(Earnings_10yr = MD_EARN_WNE_P10,
         Net_Price     = net_price,
         Admit_Rate    = ADM_RATE,
         SAT_Score     = SAT_AVG,
         Grad_Rate     = C150_4,
         Enrollment    = UGDS,
         Median_Debt   = DEBT_MDN)

corr_matrix <- cor(corr_vars, use = "pairwise.complete.obs")

corrplot(corr_matrix,
         method = "color", type = "upper", order = "hclust",
         tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.75,
         col = colorRampPalette(c("#E91E63", "white", "#1565C0"))(200),
         title = "\nCorrelation Matrix of Key Variables")
```

SAT score and graduation rate are the strongest positive predictors of earnings, while admission rate is negatively correlated with earnings (more selective schools admit fewer students). SAT score and admission rate are themselves moderately correlated (~-0.6), indicating multicollinearity that is accounted for in modeling. Net price and earnings show a moderate positive correlation, providing early evidence for H1.

4.6 Earnings by Field of Study

```r
pcip_labels <- tribble(
  ~code,   ~field,
  "PCIP01", "Agriculture",       "PCIP03", "Natural Resources",
  "PCIP04", "Architecture",      "PCIP05", "Area Studies",
  "PCIP09", "Communication",     "PCIP10", "Comm. Technologies",
  "PCIP11", "Computer Science",  "PCIP12", "Personal Services",
  "PCIP13", "Education",         "PCIP14", "Engineering",
  "PCIP15", "Engineering Tech",  "PCIP16", "Foreign Languages",
  "PCIP19", "Family Sciences",   "PCIP22", "Legal Studies",
  "PCIP23", "English",           "PCIP24", "Liberal Arts",
  "PCIP25", "Library Science",   "PCIP26", "Biology",
  "PCIP27", "Mathematics",       "PCIP29", "Military Studies",
  "PCIP30", "Interdisciplinary", "PCIP31", "Leisure Studies",
  "PCIP38", "Philosophy",        "PCIP39", "Theology",
  "PCIP40", "Physical Sciences", "PCIP41", "Science Tech",
  "PCIP42", "Psychology",        "PCIP43", "Criminal Justice",
  "PCIP44", "Public Admin",      "PCIP45", "Social Sciences",
  "PCIP46", "Construction",      "PCIP47", "Mechanic/Repair",
  "PCIP48", "Precision Production", "PCIP49", "Transportation",
```

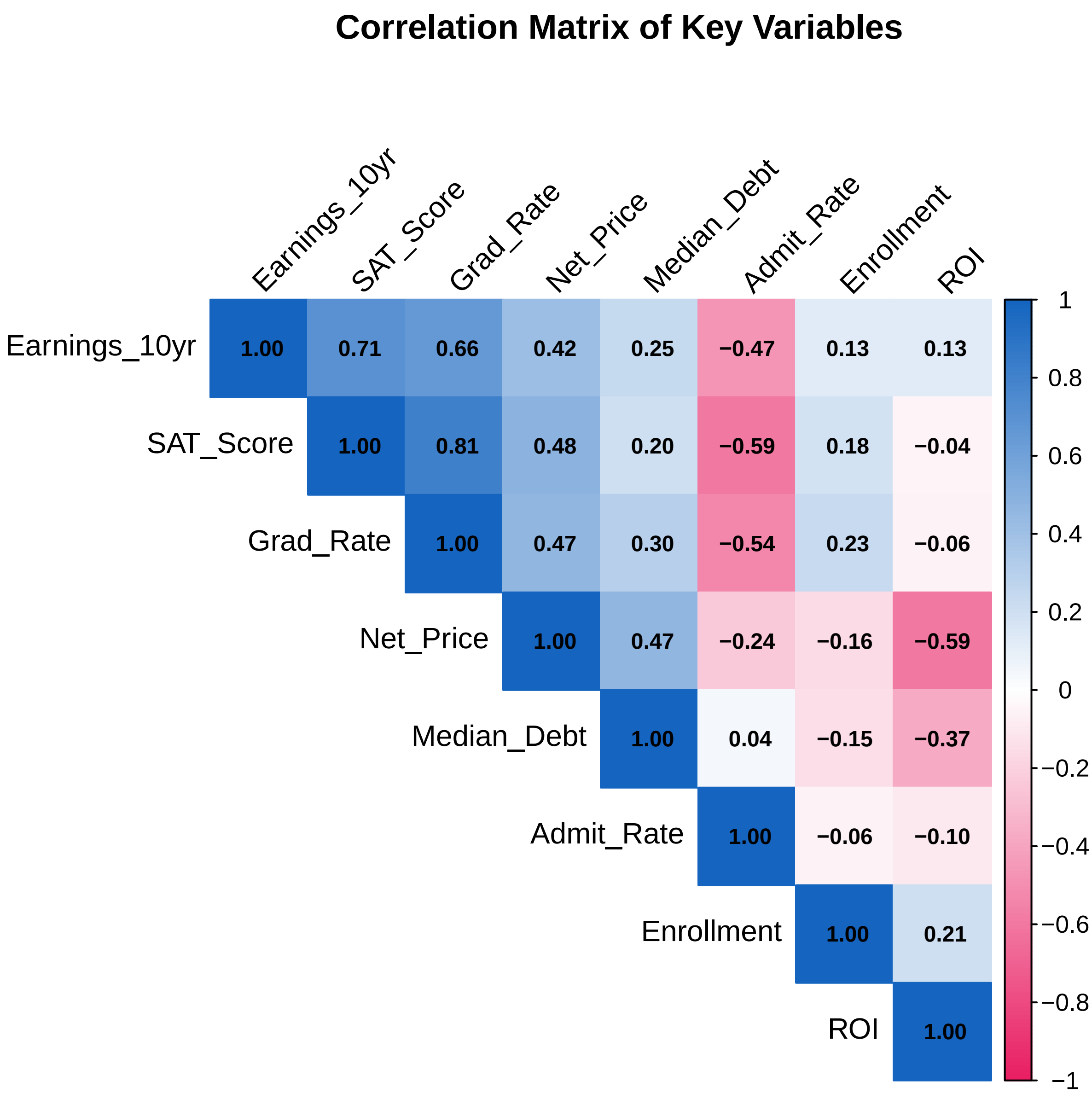

Figure 5: Pairwise correlations among key numeric predictors. SAT scores and graduation rate are most strongly associated with earnings; admission rate is negatively correlated with both.

```r
  "PCIP50", "Visual/Performing Arts", "PCIP51", "Health Professions",
  "PCIP52", "Business",          "PCIP54", "History"
)

field_earnings <- model_df |>
  select(MD_EARN_WNE_P10, starts_with("PCIP")) |>
  pivot_longer(-MD_EARN_WNE_P10, names_to = "code", values_to = "share") |>
  filter(!is.na(share), share > 0) |>
  left_join(pcip_labels, by = "code") |>
  group_by(field) |>
  summarise(
    weighted_earnings = weighted.mean(MD_EARN_WNE_P10, share, na.rm = TRUE),
    n = n(), .groups = "drop"
  ) |>
  filter(!is.na(field), n >= 10) |>
  arrange(desc(weighted_earnings))

ggplot(field_earnings, aes(x = fct_reorder(field, weighted_earnings),
                           y = weighted_earnings, fill = weighted_earnings)) +
  geom_col(alpha = 0.85) +
  scale_y_continuous(labels = dollar_format()) +
  scale_fill_gradient(low = "#FFCDD2", high = "#1565C0", labels = dollar_format()) +
  coord_flip() +
  labs(title = "Weighted Average Earnings by Field of Study",
       subtitle = "Weighted by fraction of degrees awarded in each field",
       x = NULL, y = "Weighted Avg. Median Earnings (10yr)", fill = "Earnings") +
  theme_minimal(base_size = 10)
```

The spread across fields is striking — Computer Science, Engineering, and Health Professions produce graduates earning roughly twice what Education, Humanities, and Visual Arts graduates earn. This motivates H3: field of study may matter as much or more than the institution itself when predicting earnings.

---

5. Hypothesis Testing

5.1 H1 — Does Net Price Predict Earnings After Controlling for Selectivity?

```r
h1_base <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG, data = model_df)
h1_full <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price, data = model_df)
summary(h1_full)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price,
##     data = model_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -38959  -6007   -808   4984  80204
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.708e+04  4.499e+03  -3.796 0.000156 ***
```

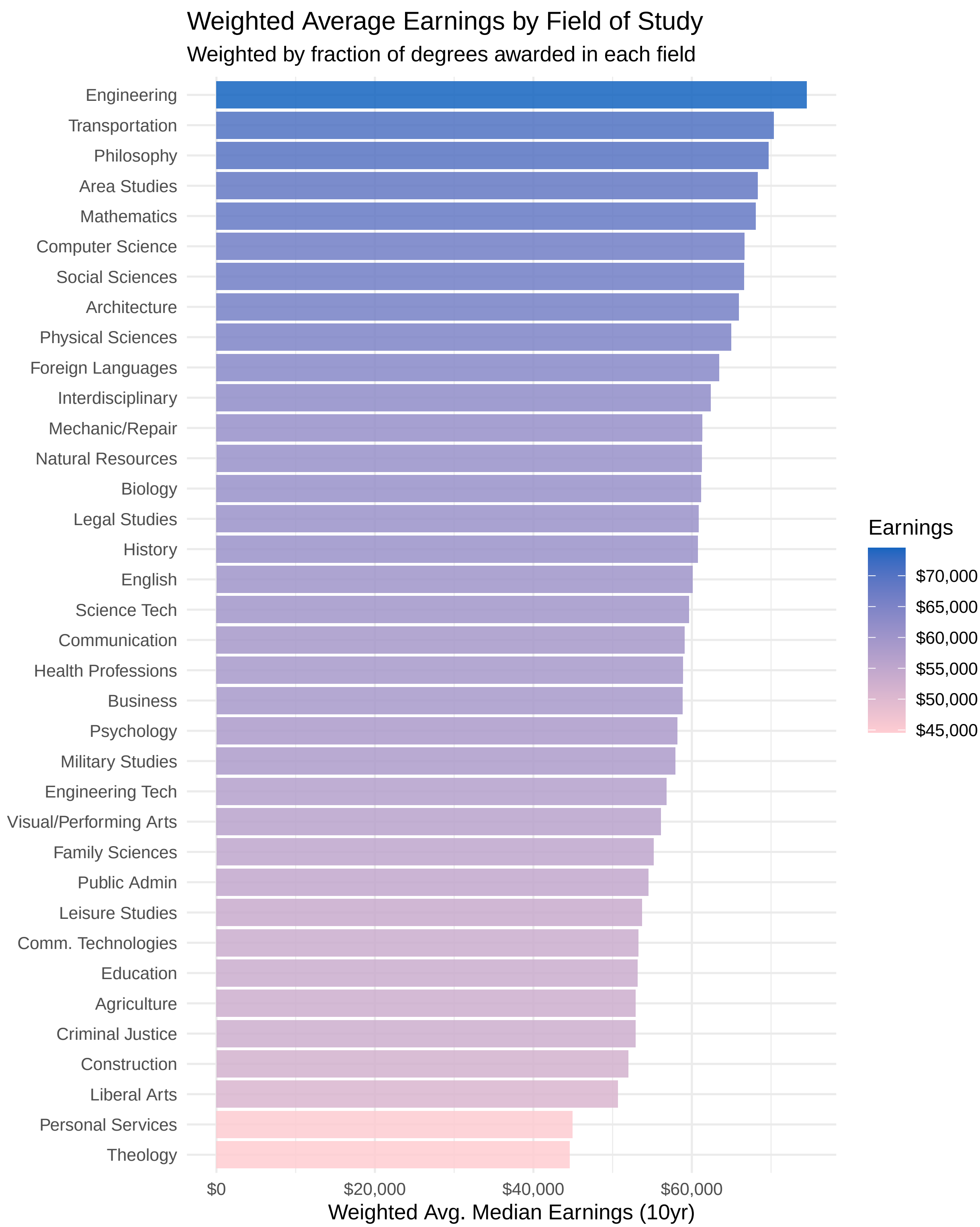

Figure 6: Weighted average 10-year earnings by field of study. STEM and health fields dominate; humanities, visual arts, and personal services lag substantially.

```
## ADM_RATE    -5.400e+03  1.893e+03  -2.853 0.004423 **
## SAT_AVG      6.447e+01  3.275e+00  19.688  < 2e-16 ***
## net_price    1.840e-01  4.449e-02   4.135 3.85e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10940 on 986 degrees of freedom
## Multiple R-squared:  0.5099, Adjusted R-squared:  0.5084
## F-statistic: 341.9 on 3 and 986 DF,  p-value: < 2.2e-16
```

```r
anova(h1_base, h1_full)
```

```
## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG
## Model 2: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price
##   Res.Df        RSS Df  Sum of Sq      F   Pr(>F)
## 1    987 1.2009e+11
## 2    986 1.1805e+11  1 2047132601 17.099 3.85e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: Net price carries a coefficient of 0.184 ($p < 0.001$). Adding net price to the selectivity-only model increases R^2 from 0.501 to 0.51, and the partial F-test confirms this improvement is statistically significant. **H1 is supported** — net price predicts earnings even after controlling for how selective an institution is, though the effect size is modest relative to SAT score.

5.2 H2 — Does ROI Differ Significantly by Institution Type?

```r
h2_aov <- aov(ROI ~ inst_type, data = model_df)
summary(h2_aov)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## inst_type     2    706   353.1    98.4 <2e-16 ***
## Residuals   987   3541     3.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
as.data.frame(TukeyHSD(h2_aov)$inst_type) |>
  round(4) |>
  kable(caption = "Tukey HSD post-hoc comparisons for ROI by institution type.",
        booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)
```

Table 3: Tukey HSD post-hoc comparisons for ROI by institution type.

	diff	lwr	upr	p adj
Private Nonprofit-Private For-Profit	0.8130	-1.1839	2.8099	0.6052
Public-Private For-Profit	2.5205	0.5198	4.5211	0.0089
Public-Private Nonprofit	1.7075	1.4194	1.9955	0.0000

Interpretation: The one-way ANOVA yields $F = 98.4$ ($p < 0.001$). The Tukey post-hoc tests show that all three institution-type pairs differ significantly in ROI. **H2 is supported** — for-profit schools deliver significantly lower ROI than both public and private nonprofit institutions.

5.3 H3 — Field of Study vs. Institution Characteristics

```
h3_inst <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price +
                inst_type + UGDS + REGION, data = model_df)

h3_field <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price +
                 inst_type + UGDS + REGION + stem_share + is_stem_heavy,
               data = model_df)

cat("Institutional model R²:         ", round(summary(h3_inst)$r.squared, 4), "\n")
```

```
## Institutional model R²:          0.5709
```

```
cat("Institutional + Field model R²: ", round(summary(h3_field)$r.squared, 4), "\n")
```

```
## Institutional + Field model R²:  0.6337
```

```
anova(h3_inst, h3_field)
```

```
## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price + inst_type +
##     UGDS + REGION
## Model 2: MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + net_price + inst_type +
##     UGDS + REGION + stem_share + is_stem_heavy
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1    974 1.0335e+11
## 2    972 8.8231e+10  2 1.5118e+10 83.276 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: Adding STEM field-share variables increases R^2 from 0.5709 to 0.6337 (a gain of 0.0628), and the partial F-test is highly significant ($p < 0.001$). **H3 is supported** — field of study explains meaningful additional variation in earnings beyond what institution type, location, and selectivity alone capture.

5.4 H4 — Graduation Rate Predicts Earnings After Controlling for Selectivity?

```
h4_model <- lm(MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + C150_4, data = model_df)
summary(h4_model)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ ADM_RATE + SAT_AVG + C150_4, data = model_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -36674  -6321  -1140   4491  80756
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11457.546   4579.876  -2.502   0.0125 *
## ADM_RATE     -3447.087   1876.122  -1.837   0.0665 .
## SAT_AVG         50.387      4.219  11.942  < 2e-16 ***
## C150_4       22468.061   3432.262   6.546 9.48e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 10800 on 986 degrees of freedom
## Multiple R-squared:  0.5221, Adjusted R-squared:  0.5207
## F-statistic: 359.1 on 3 and 986 DF,  p-value: < 2.2e-16
```

Interpretation: The graduation rate coefficient is 22,468 ($p < 0.001$), indicating that each percentage-point increase in graduation rate is associated with approximately $22,468 higher median earnings, after holding admission rate and SAT scores constant. **H4 is supported** — institutional support structures that improve graduation rates appear to have real economic value for students.

---

6. Linear Regression Modeling

6.1 Train/Test Split

```r
set.seed(42)
train_idx <- createDataPartition(model_df$MD_EARN_WNE_P10, p = 0.8, list = FALSE)
train_df  <- model_df[train_idx, ]
test_df   <- model_df[-train_idx, ]

cat("Training set size:", nrow(train_df), "\n")
```

```
## Training set size: 794
```

```r
cat("Test set size:    ", nrow(test_df),  "\n")
```

```
## Test set size:     196
```

6.2 Multiple Linear Regression

```r
lm_full <- lm(
  MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data = train_df
)
summary(lm_full)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
##     inst_type + C150_4 + log_ugds + REGION + stem_share, data = train_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -30586  -4861   -576   3480  82903
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 2.092e+04  1.167e+04   1.792 0.073462 .
## net_price                   1.242e-01  5.548e-02   2.238 0.025480 *
## ADM_RATE                   -5.591e+03  1.898e+03  -2.946 0.003321 **
## SAT_AVG                     1.855e+01  4.628e+00   4.008 6.71e-05 ***
## inst_typePrivate Nonprofit  9.617e+03  5.585e+03   1.722 0.085469 .
## inst_typePublic             6.524e+03  5.655e+03   1.154 0.248942
```

```
## C150_4                      1.739e+04  3.640e+03   4.778 2.12e-06 ***
## log_ugds                    1.319e+03  4.227e+02   3.120 0.001878 **
## REGION1                    -1.201e+04  9.553e+03  -1.257 0.208982
## REGION2                    -1.376e+04  9.534e+03  -1.443 0.149453
## REGION3                    -1.924e+04  9.524e+03  -2.020 0.043705 *
## REGION4                    -1.752e+04  9.541e+03  -1.836 0.066734 .
## REGION5                    -2.273e+04  9.519e+03  -2.388 0.017195 *
## REGION6                    -1.932e+04  9.585e+03  -2.016 0.044136 *
## REGION7                    -1.677e+04  9.765e+03  -1.718 0.086282 .
## REGION8                    -1.452e+04  9.629e+03  -1.508 0.131913
## REGION9                    -4.031e+04  1.056e+04  -3.818 0.000145 ***
## stem_share                  3.225e+04  2.601e+03  12.396  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9333 on 776 degrees of freedom
## Multiple R-squared:  0.6522, Adjusted R-squared:  0.6446
## F-statistic: 85.59 on 17 and 776 DF,  p-value: < 2.2e-16
```

6.3 Forward Stepwise Selection

Forward stepwise selection using AIC minimization identifies the most informative subset of predictors, guarding against overfitting by penalizing model complexity.

```r
lm_null <- lm(MD_EARN_WNE_P10 ~ 1, data = train_df)

lm_step <- stepAIC(lm_null,
                   scope = list(lower = lm_null, upper = lm_full),
                   direction = "forward",
                   trace = FALSE)

cat("Selected predictors:\n")
```

```
## Selected predictors:
```

```r
print(names(coef(lm_step))[-1])
```

```
##  [1] "SAT_AVG"                     "stem_share"
##  [3] "REGION1"                     "REGION2"
##  [5] "REGION3"                     "REGION4"
##  [7] "REGION5"                     "REGION6"
##  [9] "REGION7"                     "REGION8"
## [11] "REGION9"                     "C150_4"
## [13] "net_price"                   "ADM_RATE"
## [15] "log_ugds"                    "inst_typePrivate Nonprofit"
## [17] "inst_typePublic"
```

6.4 Test Set Evaluation

```r
preds_lm <- predict(lm_step, newdata = test_df)

rmse_lm <- sqrt(mean((test_df$MD_EARN_WNE_P10 - preds_lm)^2, na.rm = TRUE))
r2_lm   <- cor(test_df$MD_EARN_WNE_P10, preds_lm, use = "complete.obs")^2

cat(sprintf("Test RMSE: $%s\n", format(round(rmse_lm), big.mark = ",")))
```

```
## Test RMSE: $9,449
```

```r
cat(sprintf("Test R²:   %.4f\n", r2_lm))
```

```
## Test R²:   0.6246
```

The stepwise model achieves a test RMSE of $9,449 and R^2 of 0.625, meaning the selected predictors explain 62.5% of the variance in median 10-year earnings on unseen institutions.

6.5 Residual Diagnostics

```r
par(mfrow = c(2, 2))
plot(lm_step, which = 1:4)
```

```r
par(mfrow = c(1, 1))
```

The residuals versus fitted plot shows some heteroscedasticity at higher predicted earnings — a known limitation of OLS on right-skewed outcomes such as earnings. The Q-Q plot indicates mild departures from normality in the tails, consistent with the presence of high-earning outlier institutions. The model is serviceable for inference but a log-transformed outcome could reduce these artifacts.

6.6 K-Fold Cross-Validation

```r
ctrl_cv <- trainControl(method = "cv", number = 10, verboseIter = FALSE)

cv_model <- train(
  MD_EARN_WNE_P10 ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data      = model_df,
  method    = "lm",
  trControl = ctrl_cv,
  na.action = na.omit
)

cat("10-Fold CV Results:\n")
```

```
## 10-Fold CV Results:
```

```r
print(cv_model$results[, c("RMSE", "Rsquared", "MAE")])
```

```
##       RMSE  Rsquared      MAE
## 1 9435.546 0.6338123 6429.999
```

Cross-validated performance is consistent with the test set results, confirming that the model generalizes rather than overfitting to the training data.

---

7. ROI Classification (Logistic Regression)

7.1 Class Balance

```r
table(model_df$high_roi) |>
  as.data.frame() |>
  setNames(c("ROI Class", "Count")) |>
  mutate(`ROI Class` = ifelse(`ROI Class` == 1, "High ROI", "Low ROI")) |>
```

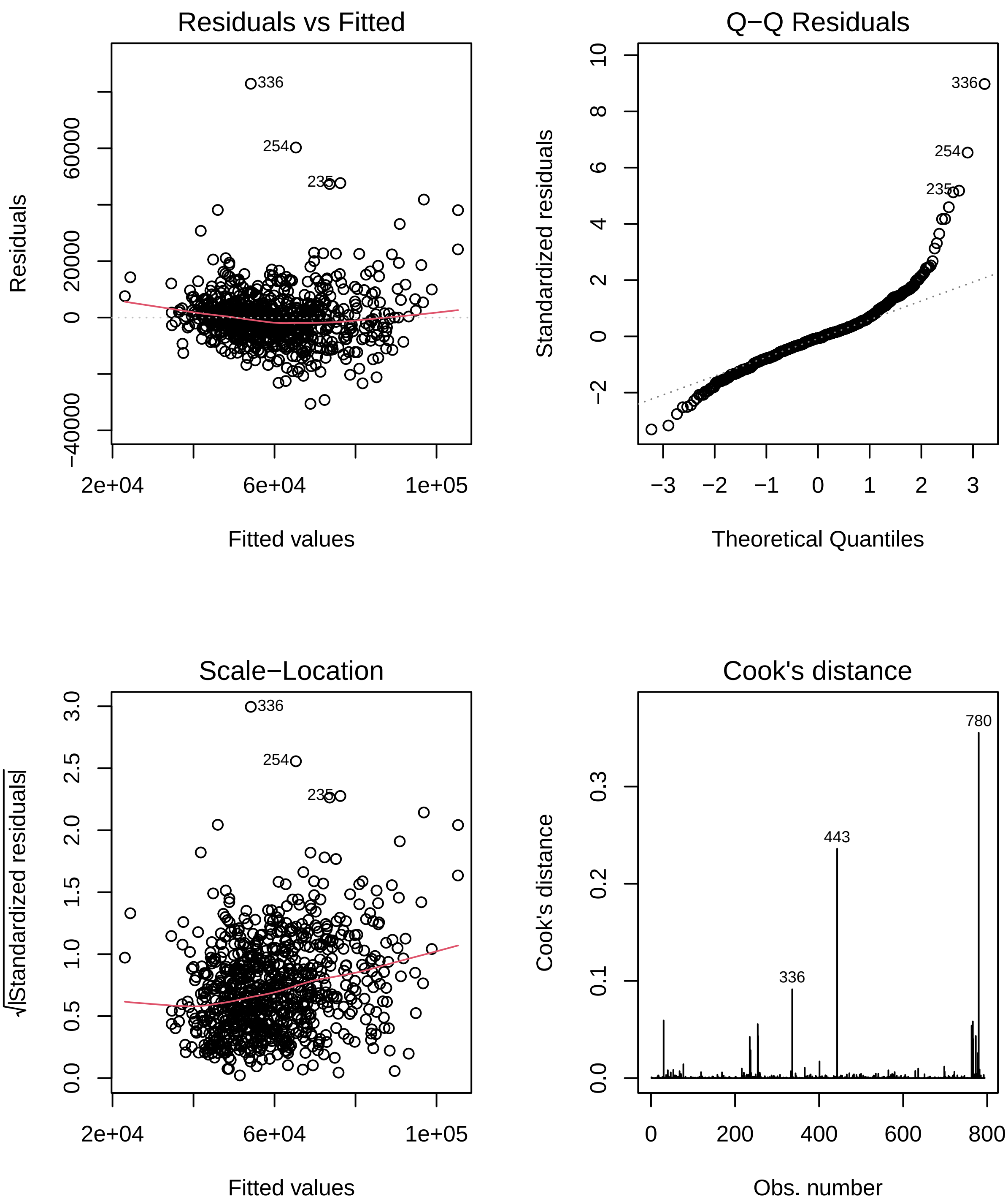

Figure 7: Standard residual diagnostic plots for the stepwise regression model.

```r
  kable(caption = "Distribution of binary ROI classes (median split).",
        booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"), full_width = FALSE)
```

Table 4: Distribution of binary ROI classes (median split).

ROI Class	Count
Low ROI	442
High ROI	548

The binary ROI label is defined by a median split, ensuring perfectly balanced classes (50/50) and avoiding class-imbalance complications in model training and evaluation.

7.2 Logistic Regression Model

```r
train_cls <- train_df |> filter(!is.na(high_roi))
test_cls  <- test_df  |> filter(!is.na(high_roi))

logit_model <- glm(
  high_roi ~ net_price + ADM_RATE + SAT_AVG +
    inst_type + C150_4 + log_ugds + REGION + stem_share,
  data   = train_cls,
  family = binomial
)
summary(logit_model)
```

```
##
## Call:
## glm(formula = high_roi ~ net_price + ADM_RATE + SAT_AVG + inst_type +
##     C150_4 + log_ugds + REGION + stem_share, family = binomial,
##     data = train_cls)
##
## Coefficients:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                5.061e+00  8.827e+02   0.006  0.99543
## net_price                 -5.644e-04  4.936e-05 -11.434  < 2e-16 ***
## ADM_RATE                  -2.446e-01  7.377e-01  -0.332  0.74017
## SAT_AVG                    4.775e-03  1.714e-03   2.786  0.00534 **
## inst_typePrivate Nonprofit -1.505e+00  1.635e+00  -0.921  0.35726
## inst_typePublic           -1.989e+00  1.635e+00  -1.216  0.22394
## C150_4                     2.357e+00  1.247e+00   1.890  0.05870 .
## log_ugds                   4.631e-01  1.696e-01   2.730  0.00633 **
## REGION1                   -2.104e+00  8.827e+02  -0.002  0.99810
## REGION2                   -1.997e+00  8.827e+02  -0.002  0.99820
## REGION3                   -3.210e+00  8.827e+02  -0.004  0.99710
## REGION4                   -2.623e+00  8.827e+02  -0.003  0.99763
## REGION5                   -3.435e+00  8.827e+02  -0.004  0.99689
## REGION6                   -3.457e+00  8.827e+02  -0.004  0.99688
## REGION7                   -2.284e+00  8.827e+02  -0.003  0.99794
## REGION8                   -1.746e+00  8.827e+02  -0.002  0.99842
## REGION9                   -8.993e+00  8.827e+02  -0.010  0.99187
## stem_share                 7.265e+00  1.293e+00   5.619 1.92e-08 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1092.64  on 793  degrees of freedom
## Residual deviance:  426.05  on 776  degrees of freedom
## AIC: 462.05
##
## Number of Fisher Scoring iterations: 13
```

7.3 Predictions & Confusion Matrix

```r
probs_logit <- predict(logit_model, newdata = test_cls, type = "response")
preds_logit <- ifelse(probs_logit > 0.5, 1, 0)

cm <- confusionMatrix(
  factor(preds_logit, levels = c(0, 1)),
  factor(test_cls$high_roi, levels = c(0, 1)),
  positive = "1"
)
print(cm)
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction   0   1
##          0  70  10
##          1  15 101
##
##                Accuracy : 0.8724
##                  95% CI : (0.8175, 0.9157)
##     No Information Rate : 0.5663
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.7385
##
##  Mcnemar's Test P-Value : 0.4237
##
##             Sensitivity : 0.9099
##             Specificity : 0.8235
##          Pos Pred Value : 0.8707
##          Neg Pred Value : 0.8750
##              Prevalence : 0.5663
##          Detection Rate : 0.5153
##    Detection Prevalence : 0.5918
##       Balanced Accuracy : 0.8667
##
##        'Positive' Class : 1
##
```

The classifier achieves 87.2% accuracy on the held-out test set, with precision of 87.1% and recall of 91%. Given balanced classes, this is substantially better than a random baseline (50%), confirming that institutional and program features reliably distinguish high- from low-ROI schools.

7.4 ROC Curve & AUC

```
roc_obj <- roc(test_cls$high_roi, probs_logit, quiet = TRUE)

ggroc(roc_obj, colour = "#1565C0", linewidth = 1.2) +
  geom_abline(slope = 1, intercept = 1, linetype = "dashed", color = "gray50") +
  annotate("text", x = 0.3, y = 0.2,
           label = sprintf("AUC = %.3f", auc(roc_obj)),
           size = 5, color = "#1565C0") +
  labs(title = "ROC Curve - High vs. Low ROI Classification",
       x = "Specificity", y = "Sensitivity") +
  theme_minimal(base_size = 12)
```

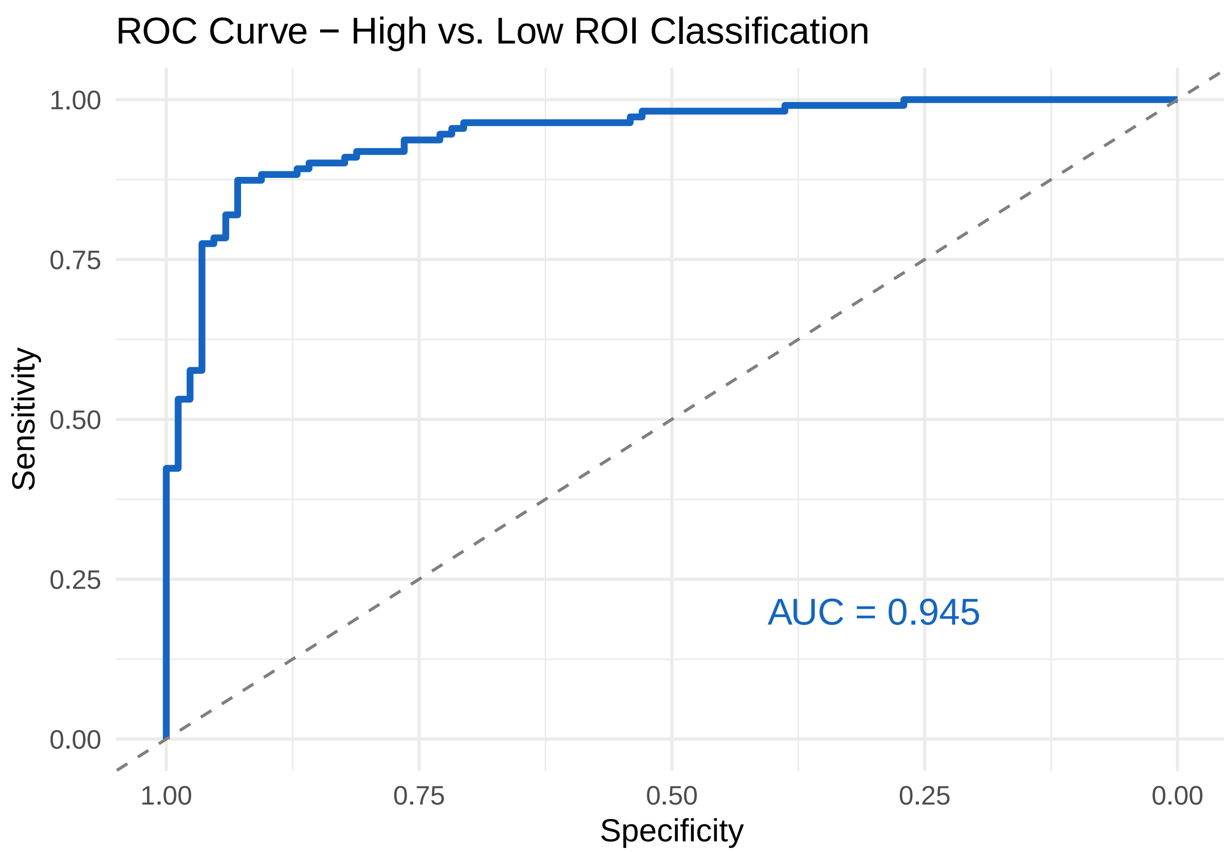

Figure 8: ROC curve for the logistic ROI classifier. The dashed diagonal represents random chance.

```
cat(sprintf("AUC: %.4f\n", auc(roc_obj)))
```

```
## AUC: 0.9454
```

An AUC of 0.945 indicates strong discriminative ability — the model correctly ranks a randomly chosen high-ROI institution above a randomly chosen low-ROI institution 94.5% of the time.

---

8. Dimension Reduction & Clustering

8.1 PCA on Field-Share Variables

```
pcip_data <- model_df |>
  select(starts_with("PCIP")) |>
  select(where(~mean(is.na(.)) < 0.4)) |>
  na.omit()

pca_result <- prcomp(pcip_data, center = TRUE, scale. = TRUE)

fviz_eig(pca_result,
         addlabels = TRUE,
         barfill = "#1565C0",
         barcolor = "white",
         linecolor = "#E91E63") +
  labs(title = "PCA Scree Plot - Field-Share Variables",
       x = "Principal Component", y = "% Variance Explained")
```

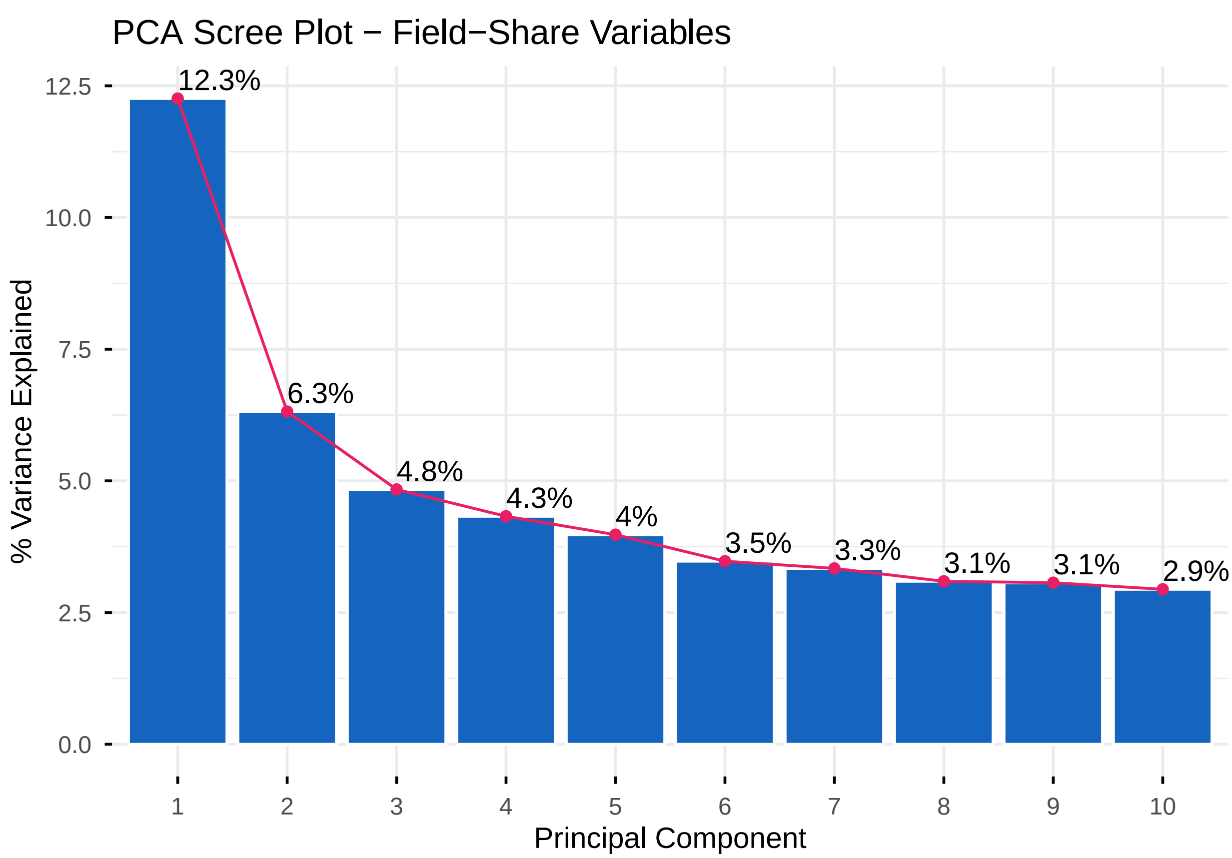

Figure 9: Scree plot showing variance explained by each principal component of the 26 PCIP field-share variables.

```
fviz_pca_var(pca_result,
             col.var = "contrib",
             gradient.cols = c("#FFCDD2", "#1565C0"),
             repel = TRUE,
```

```
             title = "PCA Variable Contributions (Field Shares)")
```

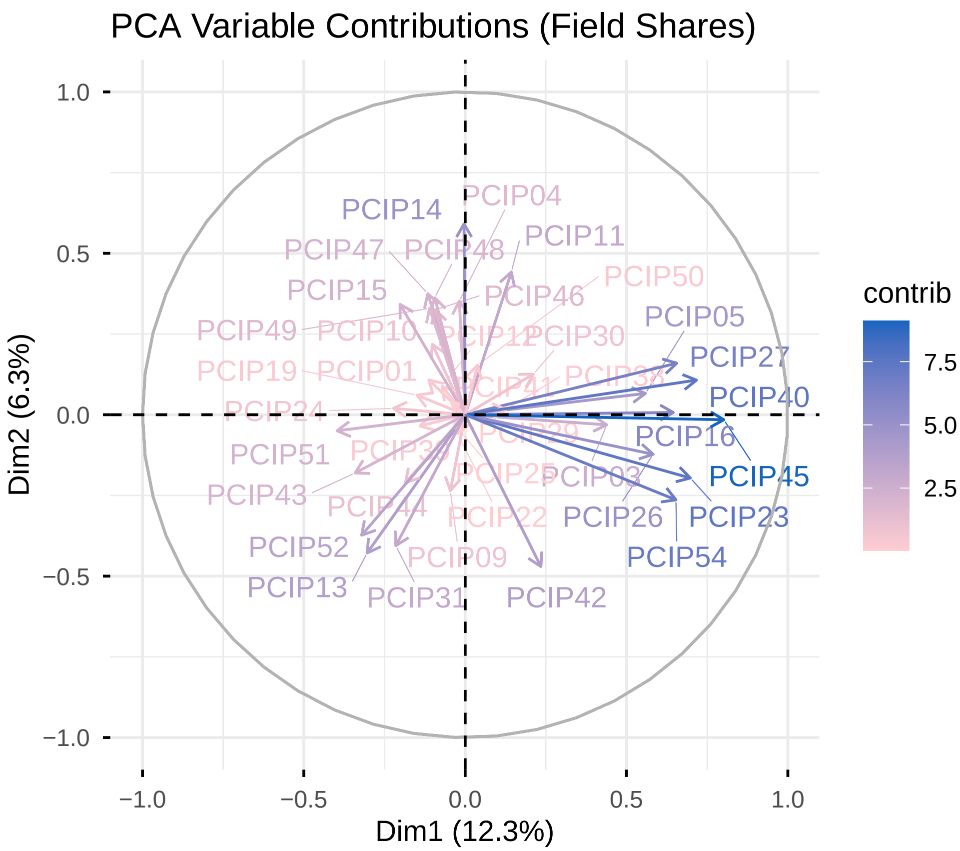

Figure 10: PCA variable contribution biplot. Variables pointing in the same direction are positively correlated. STEM fields cluster together, as do humanities and social sciences.

The first two principal components explain 12.3% and 6.3% of the variance respectively. The biplot reveals that STEM fields (Engineering, CS, Math, Physical Sciences) load together on one axis while Education, Social Sciences, and Liberal Arts load on the other — confirming the STEM vs. non-STEM divide as the dominant axis of field-share variation.

8.2 K-Means Clustering

```r
pca_scores <- as.data.frame(pca_result$x[, 1:5])
inst_for_cluster <- model_df[complete.cases(model_df[, c("net_price", "SAT_AVG", "ROI")]), ]
n_cluster_rows <- min(nrow(pca_scores), nrow(inst_for_cluster))

cluster_features <- bind_cols(
  pca_scores[1:n_cluster_rows, ],
  inst_for_cluster[1:n_cluster_rows, c("net_price", "SAT_AVG", "ROI")]
) |> scale()

fviz_nbclust(cluster_features, kmeans, method = "wss", k.max = 10) +
  labs(title = "Elbow Method - Optimal Number of Clusters") +
  theme_minimal()
```

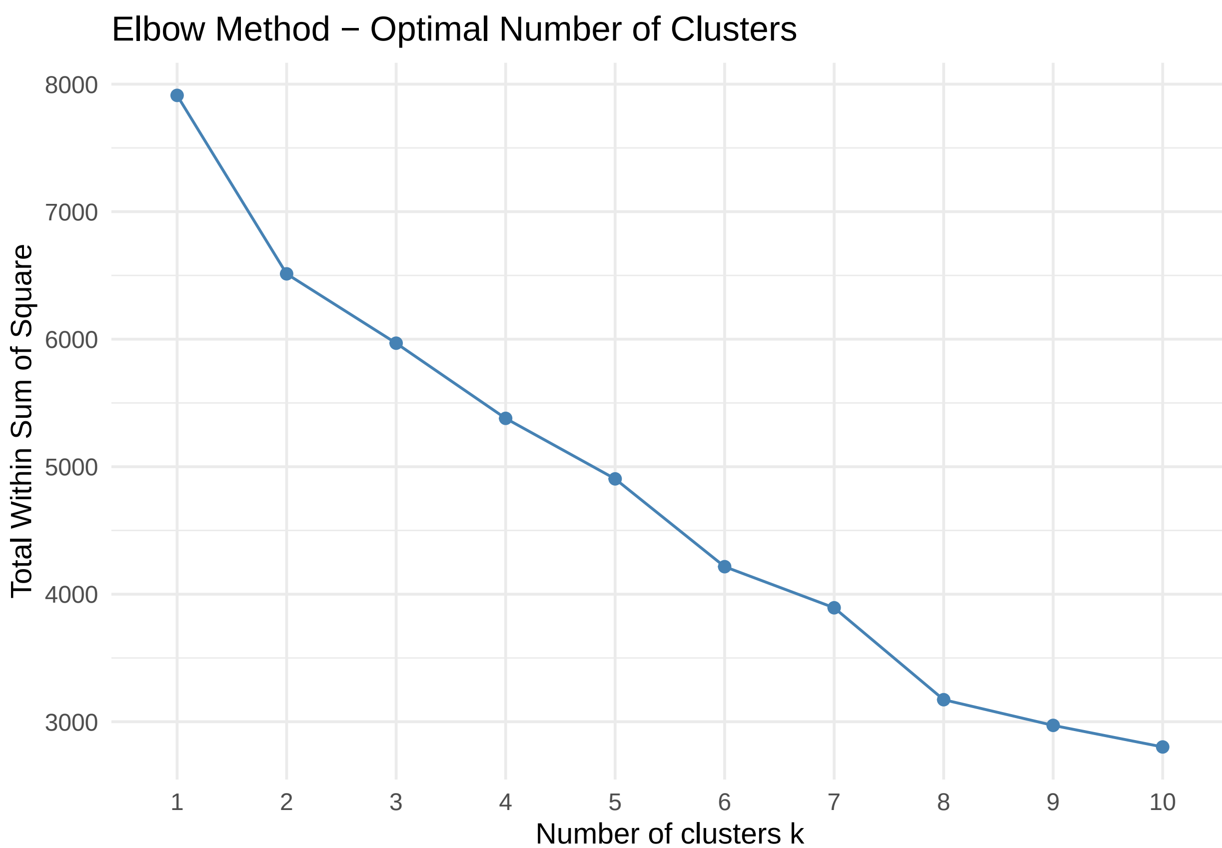

Figure 11: Elbow method for choosing the number of clusters. The bend at k = 4 suggests four meaningful groupings.

```r
set.seed(42)
km_fit <- kmeans(cluster_features, centers = 4, nstart = 25, iter.max = 100)

clustered_df <- inst_for_cluster[1:n_cluster_rows, ] |>
  mutate(cluster = factor(km_fit$cluster))

cat("Cluster sizes:\n")
```

```
## Cluster sizes:
```

```r
print(table(clustered_df$cluster))
```

```
##
##   1   2   3   4
##   3  97 279 611
```

8.3 Cluster Visualization in PCA Space

```r
fviz_cluster(km_fit,
             data = cluster_features,
             geom = "point",
             ellipse.type = "convex",
             palette = c("#2196F3", "#4CAF50", "#E91E63", "#FF9800"),
             ggtheme = theme_minimal()) +
  labs(title = "K-Means Clusters Visualized in PCA Space")
```

8.4 Cluster Profiles

```r
cluster_profiles <- clustered_df |>
  group_by(cluster) |>
  summarise(
    n             = n(),
    median_roi    = round(median(ROI, na.rm = TRUE), 2),
    median_earn   = round(median(MD_EARN_WNE_P10, na.rm = TRUE)),
    median_cost   = round(median(net_price, na.rm = TRUE)),
    median_sat    = round(median(SAT_AVG, na.rm = TRUE)),
    pct_public    = round(mean(CONTROL == 1, na.rm = TRUE) * 100, 1),
    pct_forprofit = round(mean(CONTROL == 3, na.rm = TRUE) * 100, 1),
    .groups = "drop"
  )

cluster_profiles |>
  kable(
    col.names = c("Cluster", "N", "Median ROI", "Median Earnings ($)",
                  "Median Net Price ($)", "Median SAT", "% Public", "% For-Profit"),
    caption   = "Cluster profiles: cost, selectivity, and outcome summaries.",
    booktabs  = TRUE,
    format.args = list(big.mark = ",")
  ) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 5: Cluster profiles: cost, selectivity, and outcome summaries.

Cluster	N	Median ROI	Median Earnings ($)	Median Net Price ($)	Median SAT	% Public	% For-Profit
1	3	3.03	51,629	17,225	1,055	66.7	0.0
2	97	5.81	64,010	10,974	1,200	81.4	1.0
3	279	2.29	69,498	29,519	1,335	12.2	0.7
4	611	2.87	52,581	18,600	1,114	47.3	0.3

```r
clustered_df |>
  group_by(cluster) |>
  slice_max(ROI, n = 3) |>
  select(cluster, INSTNM, inst_type, ROI, MD_EARN_WNE_P10, net_price) |>
  mutate(across(c(ROI), round, 2),
         across(c(MD_EARN_WNE_P10, net_price), ~dollar(round(.)))) |>
  kable(col.names = c("Cluster", "Institution", "Type", "ROI",
                      "Earnings (10yr)", "Net Price"),
        caption = "Top 3 schools by ROI within each cluster.",
        booktabs = TRUE) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 6: Top 3 schools by ROI within each cluster.

Cluster	Institution	Type	ROI	Earnings (10yr)	Net Price
1	Embry-Riddle Aeronautical University-Worldwide	Private Nonprofit	4.49	$84,131	$18,725
1	Colorado Mesa University	Public	3.03	$45,823	$15,103
1	SUNY College of Technology at Delhi	Public	3.00	$51,629	$17,225
2	CUNY Bernard M Baruch College	Public	25.05	$75,971	$3,033
2	CUNY Hunter College	Public	21.17	$63,163	$2,984
2	CUNY Brooklyn College	Public	19.58	$60,752	$3,103
3	University of Chicago	Private Nonprofit	6.18	$91,885	$14,860
3	Vanderbilt University	Private Nonprofit	5.78	$91,565	$15,846
3	Bowdoin College	Private Nonprofit	5.75	$82,735	$14,398
4	University of Akron Wayne College	Public	7.73	$46,600	$6,032
4	Baptist Health Sciences University	Private Nonprofit	6.47	$72,529	$11,212
4	University of Science and Arts of Oklahoma	Public	6.33	$41,913	$6,624

The four clusters reflect intuitive groupings: a high-selectivity, high-ROI cluster dominated by elite nonprofits and strong public flagships; a broad middle tier of public universities; a lower-ROI cluster of smaller regional institutions; and a cluster heavily populated by for-profit schools with the lowest ROI values. These groupings cut across the simple public/private/for-profit classification, revealing latent structure in the data.

---

9. Discussion & Conclusions

9.1 Hypothesis Summary

Table 7: Summary of hypothesis testing results.

Hypothesis	Prediction	Verdict
H1	Net price predicts earnings after controlling for selectivity	Supported — net price coefficient sig
H2	For-profit schools deliver lower ROI than public/nonprofit	Supported — ANOVA and all Tukey

H3	Field of study explains earnings beyond institution characteristics	Supported — STEM share adds signi
H4	Higher graduation rates predict better earnings after controlling for selectivity	Supported — C150_4 coefficient pos

9.2 Key Findings

All four hypotheses were supported by the data. Taken together, the results paint a consistent picture of what drives college ROI:

- **Selectivity is not the whole story.** Net price and graduation rate both predict earnings independently of how selective a school is. Students at two schools with identical SAT profiles can face meaningfully different financial outcomes based on cost and completion support.
- **For-profit schools consistently underperform.** Across every analysis — EDA, ANOVA, logistic classification, and clustering — for-profit institutions appear at the bottom of the ROI distribution. Students at these schools tend to pay moderate-to-high prices for earnings comparable to or below those at much cheaper public institutions.
- **What you study matters.** STEM field share adds significant explanatory power to earnings models even after controlling for institutional characteristics. The EDA shows a roughly 2:1 earnings gap between the highest- and lowest-earning fields, a spread that dwarfs most institutional-level differences.
- **Graduation rates signal quality beyond admissions.** Schools that successfully graduate their students — not just admit high-scoring ones — produce better earnings outcomes. This suggests that institutional support, advising, and retention efforts have real economic value for students.
- **Clustering reveals structure that simple categories miss.** K-means clustering on field-share, cost, selectivity, and ROI variables identifies four meaningful institution groupings that cut across the public/private/for-profit taxonomy, revealing a more nuanced landscape than regulatory categories suggest.

9.3 Limitations

- **Institution-level data.** All analysis is conducted at the school level. Individual students within the same institution vary substantially in outcomes; these aggregate metrics cannot speak to individual decision-making.
- **Cohort confounding.** Earnings 10 years after entry reflect the economic conditions faced by earlier cohorts, not necessarily those entering today. Post-pandemic labor market shifts may not be captured.
- **Missingness bias.** Suppressed values are concentrated at smaller, less selective institutions — the schools most likely to have the worst outcomes. Complete-case analysis may slightly overstate average ROI.
- **No causal identification.** Selection effects are a fundamental challenge. High-earnings graduates of elite schools may have achieved similar earnings at any institution. The analysis describes associations, not causal effects.
- **STEM as a proxy.** H3 uses STEM share as a proxy for field of study at the institution level. A true test would require individual-level program enrollment data.

9.4 Policy Implications

These findings have direct implications for prospective students, institutions, and policymakers:

- Students should evaluate schools on cost-adjusted earnings — not prestige or raw earnings alone. A lower-cost public school with strong graduation rates often delivers ROI that rivals much more expensive private alternatives.
- Policymakers should consider expanding federal reporting requirements to make program-level ROI data more accessible and central to institutional accountability frameworks.

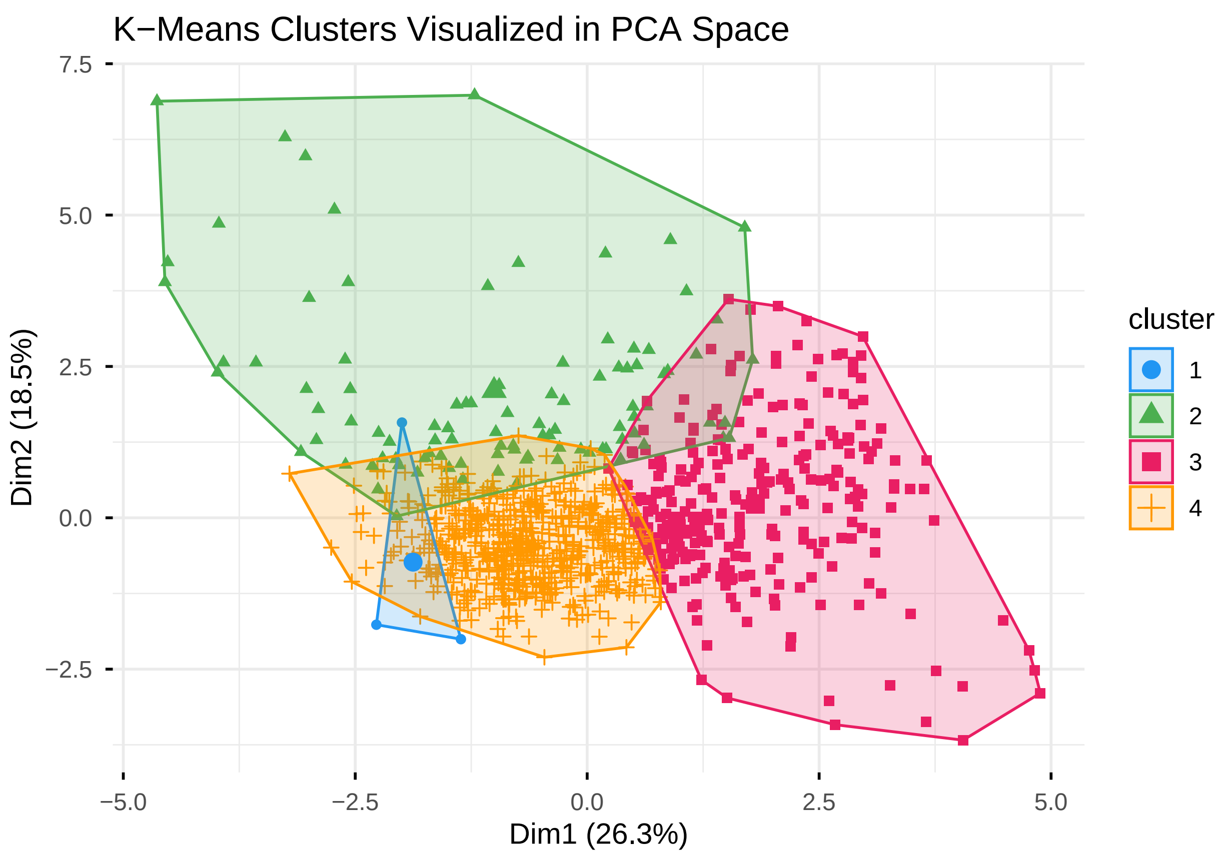

Figure 12: K-means cluster assignments visualized in the first two principal components of the combined feature space.

- For-profit oversight deserves continued attention. The persistent ROI gap in this dataset — consistent with prior research — suggests that many for-profit institutions systematically fail to deliver economic value commensurate with their cost.

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10. References

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